E · Steel rail fabrication and drilling

All steel work happens on the bench, before either rail goes near a top. Under the desk you only drive screws and bolts; you never drill steel there. Two identical lengths of 2 × 2 × 3/16 in hot-rolled angle arrive; either can be rail A. Tape readings in cm, caliper and drill sizes in mm.

MEASURE BEFORE MARKING OR DRILLING MEASURE ON HARDWARE
1 · Tape each rail from end to end and write the length on the tape label. Design length **213 cm**; bought as 84 in (2133.6 mm, so about 213.4 cm). If a rail is under 213 cm, stop and check before drilling.
2 · Calipers on each angle: legs 50.8 mm, thickness 4.76 mm (the docs expect about 4.8 mm).
3 · Every station is measured from the **labelled end**: rail A's **wall end**, rail B's **corner end**. The **far end carries any extra** length. Never centre the hole pattern on the bar and never shift a hole to share out the extra.

1 **Label the datum ends.** Wrap tape round one end of each rail and write on it: **RAIL A · WALL END** on one, **RAIL B · CORNER END** on the other. Measure both rails from those ends (Fig. E-1). File both cut ends of each rail smooth.

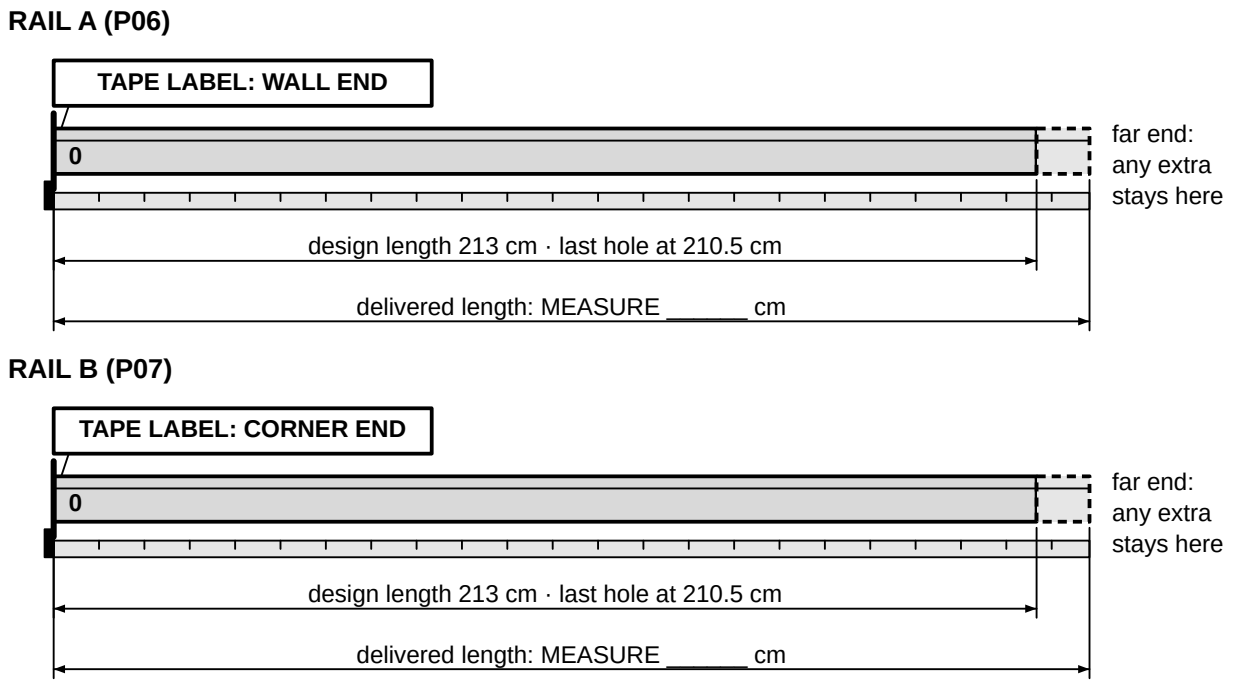


Fig. E-1 · The labelled end is 0 for every station. The last hole is at 210.5 cm on both rails, so a longer rail only lengthens the far end beyond it. SOURCE-DEFINED design length; MEASURE ON HARDWARE delivered length.

You need	For
P06 rail A, P07 rail B; one P08 fitting as the template	the work pieces
P18 printed tools: edge T-square, rail marking gauge, rail drill jig (PETG), 2 fitting slot plugs	stations, the 19 and 29 lines, pilots, bolt-hole transfer
Bits 1/8, 7/32 and 7/16 in (the 7/16 cobalt), cutting oil, flat file, 2 clamps, tape, combination square, pencil	drilling and deburring; a drill press if you have one
Acetone or mineral spirits, 120 grit, flat black enamel	finish (step 9)

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Which way up: the section and the two lines

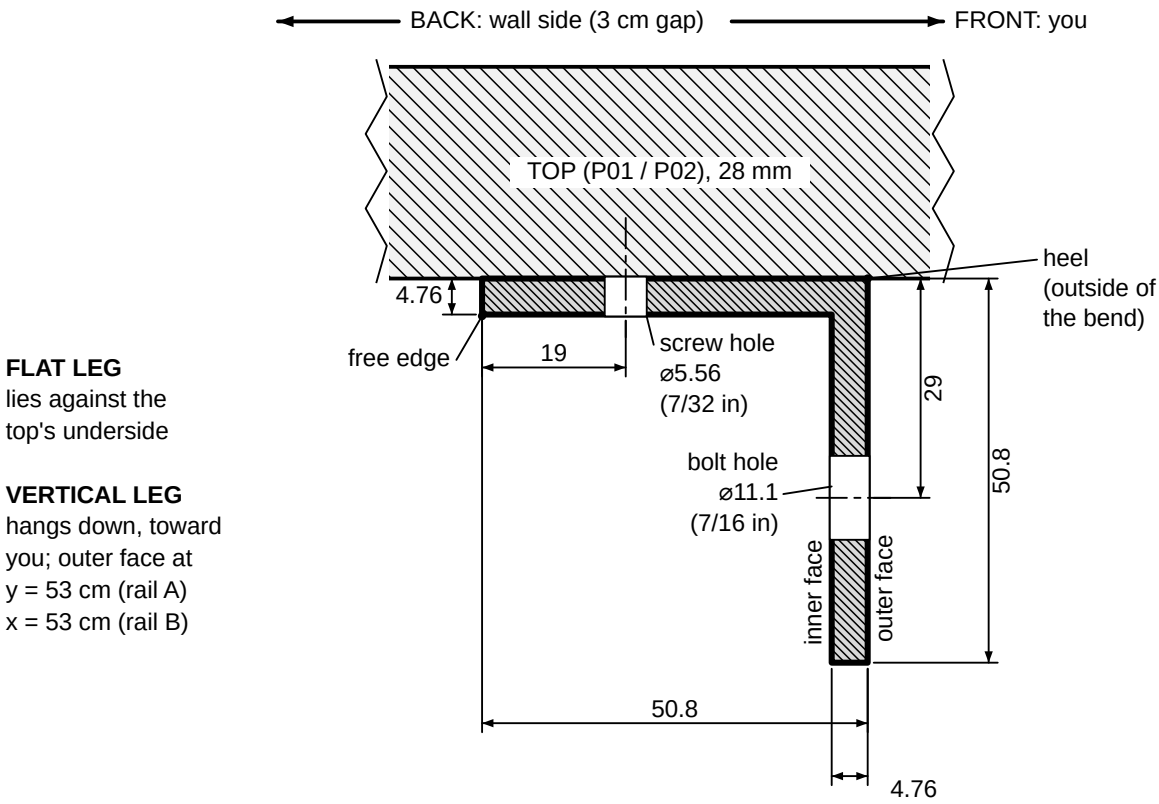


Fig. E-2 · Rail A seen end-on from its far end, as it hangs under top A (rail B is the same section turned 90°). Datum: top A's back-left corner, x along the back wall, y along the side wall. The vertical leg's outer face is at y = 53 cm on rail A and x = 53 cm on rail B, 12 cm back from the front edge (65 cm). Section dimensions in mm.

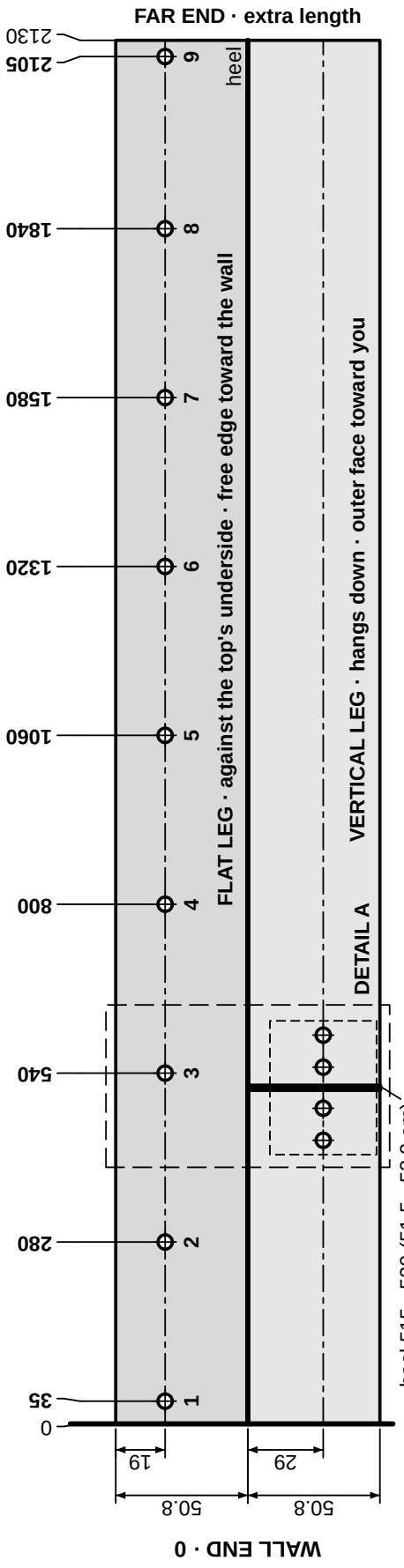
Item	mm	What it means	Confidence
Section	50.8 × 50.8 × 4.76	2 × 2 × 3/16 in hot-rolled angle	SOURCE-DEFINED MEASURE ON HARDWARE thickness
Flat leg		lies against the top's underside; its free edge points to the wall	SOURCE-DEFINED
Vertical leg		hangs down on the side nearer you; outer face y = 53 cm (A), x = 53 cm (B)	SOURCE-DEFINED
Screw line	19	from the flat leg's free edge; centres of the ⌀5.56 (7/32 in) screw holes	SOURCE-DEFINED
Bolt line	29	below the heel (the top's underside); centres of the ⌀11.1 (7/16 in) bolt holes	SOURCE-DEFINED
Fitting slots	21 × 12	two per P08 leg, centres 29.15 and 78.45 from the heel; legs 100.5 × 41, 5 thick (all mm)	VERIFIED 23 Oct 2026

Bench set-up. Clamp the rail with the flat leg's face up and the vertical leg hanging down over the bench edge. That is how it is marked and how the flat-leg holes are drilled. For the vertical leg (steps 6 and 7) turn it so the vertical leg's outer face is up and the flat leg hangs over the edge.

The two fabrication drawings that follow are full pages, turned sideways: turn the booklet a quarter turn clockwise to read them. Every hole is given from the labelled end, never from the hole before: the drawings carry mm ordinates, the hole table beside each drawing gives the same stations in cm for the tape.

P06 · RAIL A · OPENED-OUT FABRICATION DRAWING · ordinates in mm (cm in the table)

Lengths 1:10 from the WALL END (datum 0); leg widths 1:2.5, not to scale across; hole symbols not to scale. Seen from outside the L.



9× ø5.56 (7/32 in) THRU in the flat leg, centres on the screw line 19 from the free edge, at the ordinates above.
4× ø11.1 (7/16 in) THRU in the vertical leg on the bolt line 29 below the heel, at 43.7, 48.6, 54.9, 59.8 cm (Detail A, mm).
Positions VERIFIED from the measured fitting; drill them THROUGH the clamped fitting with the slot plug (step 6).

Holes from the wall end

No.	mm	cm	hole	leg, line
1	35	3.5	ø5.56 (7/32 in)	flat, 19 line
2	280	28	ø5.56 (7/32 in)	flat, 19 line
3	540	54	ø5.56 (7/32 in)	flat, 19 line
4	800	80	ø5.56 (7/32 in)	flat, 19 line
5	1060	106	ø5.56 (7/32 in)	flat, 19 line
6	1320	132	ø5.56 (7/32 in)	flat, 19 line
7	1580	158	ø5.56 (7/32 in)	flat, 19 line
8	1840	184	ø5.56 (7/32 in)	flat, 19 line
9	2105	210.5	ø5.56 (7/32 in)	flat, 19 line
I2	436.6	43.7	ø11.1 (7/16 in)	vertical, 29
I1	485.9	48.6	ø11.1 (7/16 in)	vertical, 29
O1	549.2	54.9	ø11.1 (7/16 in)	vertical, 29
O2	598.5	59.8	ø11.1 (7/16 in)	vertical, 29

P06 Rail A · 2 × 2 × 3/16 in hot-rolled angle
50.8 × 50.8 × 4.76 mm, 213 cm: MEASURE first
On the desk: wall end x = 1 cm, outer face y = 53 cm
(datum: top A's back-left corner)

DETAIL A · 1:2 · dashed: the fitting clamped as the template

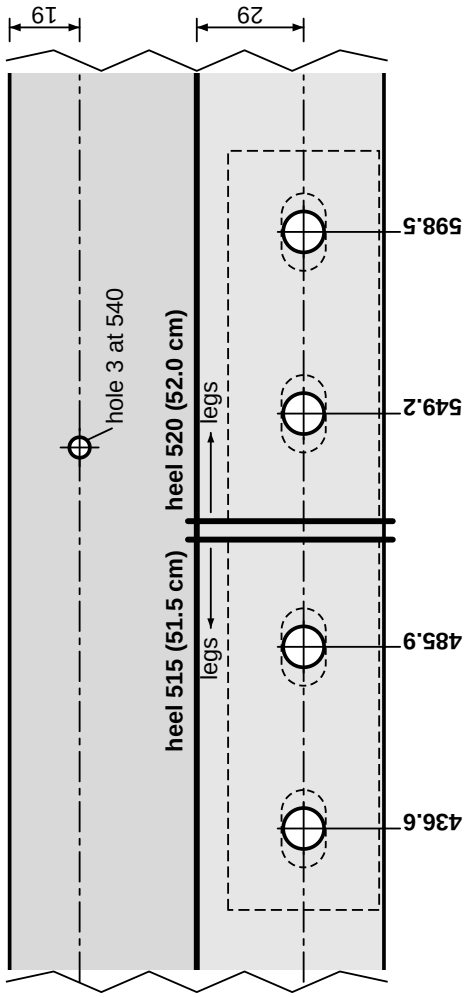
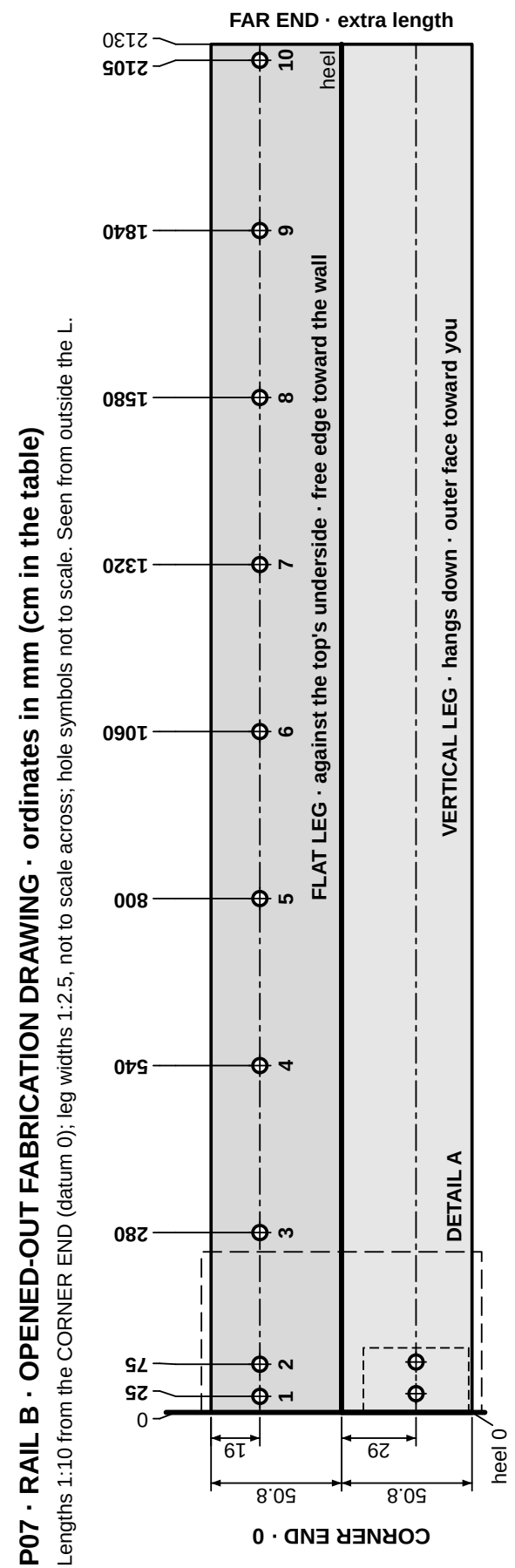


Fig. E-3 · P06 rail A, opened out: the flat leg (upper strip) and the vertical leg (lower strip) unfolded about the heel. Nine screw holes from the wall end; four bolt holes at 43.7, 48.6, 54.9 and 59.8 cm, drilled through the fitting (heels on the 51.5 and 52.0 cm marks). Ordinates in mm, cm in the table. SOURCE-DEFINED stations; VERIFIED bolt holes.



10× ø5.56 (7/32 in) THRU in the flat leg, centres on the screw line 19 from the free edge, at the ordinates above.
2× ø11.1 (7/16 in) THRU in the vertical leg on the bolt line 29 below the heel, at 2.9, 7.8 cm (Detail A, mm).
Positions VERIFIED from the measured fitting; drill them THROUGH the clamped fitting with the slot plug (step 6).

Holes from the corner end

No.	mm	cm	hole	leg, line
1	25	2.5	ø5.56 (7/32 in)	flat, 19 line
2	75	7.5	ø5.56 (7/32 in)	flat, 19 line
3	280	28	ø5.56 (7/32 in)	flat, 19 line
4	540	54	ø5.56 (7/32 in)	flat, 19 line
5	800	80	ø5.56 (7/32 in)	flat, 19 line
6	1060	106	ø5.56 (7/32 in)	flat, 19 line
7	1320	132	ø5.56 (7/32 in)	flat, 19 line
8	1580	158	ø5.56 (7/32 in)	flat, 19 line
9	1840	184	ø5.56 (7/32 in)	flat, 19 line
10	2105	210.5	ø5.56 (7/32 in)	flat, 19 line
C1	29.2	2.9	ø11.1 (7/16 in)	vertical, 29
C2	78.5	7.8	ø11.1 (7/16 in)	vertical, 29

P07 Rail B · 2 x 2 x 3/16 in hot-rolled angle
50.8 × 50.8 × 4.76 mm, 213 cm: MEASURE first
On the desk: corner end y = 53 cm, outer face x = 53 cm
(datum: top A's back-left corner)

DETAIL A · 1:2 · dashed: the fitting clamped as the template
120 (12 cm): under top A
top A's front edge (the joint)

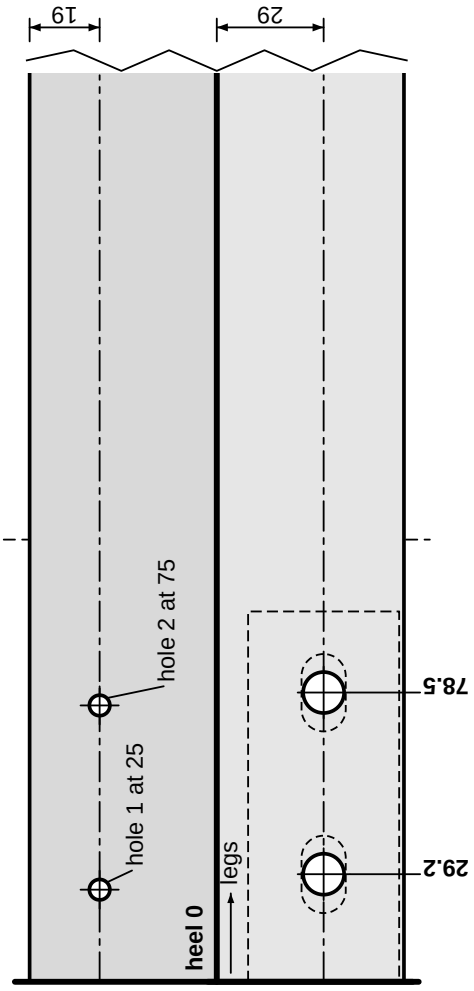
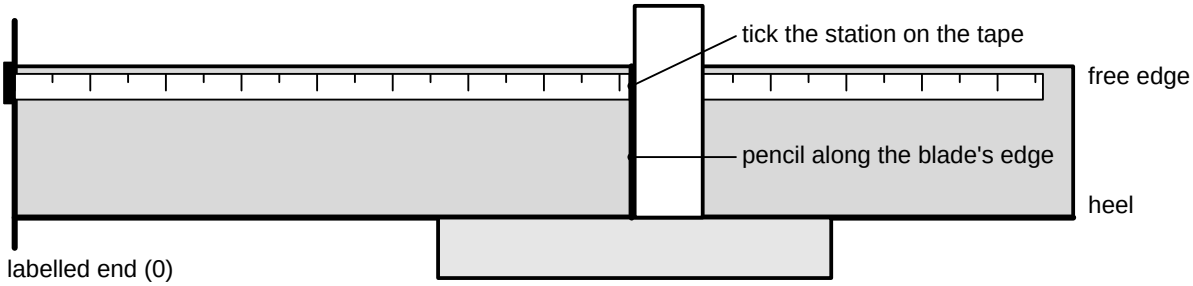


Fig. E-4 · P07 rail B, opened out. Ten screw holes from the corner end: holes 1 and 2 go into top A (the first 12 cm sits under top A), holes 3 to 10 into top B. Two bolt holes at 2.9 and 7.8 cm, shared by both fittings, drilled through the fitting with its heel flush with the corner end. SOURCE-DEFINED stations; VERIFIED bolt holes.

2 **Mark the stations.** Hook the tape over the labelled end and tick every station on the flat leg from that one hook: rail A **3.5, 28, 54, 80, 106, 132, 158, 184, 210.5 cm**; rail B **2.5, 7.5, 28, 54, 80, 106, 132, 158, 184, 210.5 cm**. Square each tick across the leg along the T-square's blade, its lip hooked over the heel edge.

Hook the tape over the labelled end once and tick every station from that one hook;
never measure from one hole to the next.



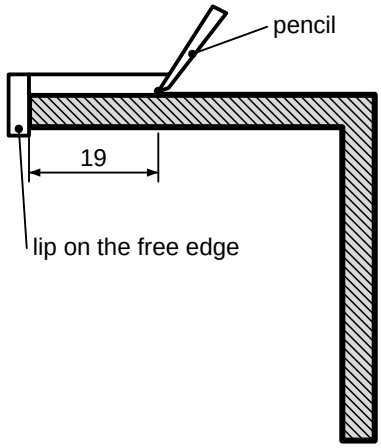
T-square (P18): lip over the heel edge, blade across the leg

Plan: flat leg face up, vertical leg over the bench edge, labelled end at the left.

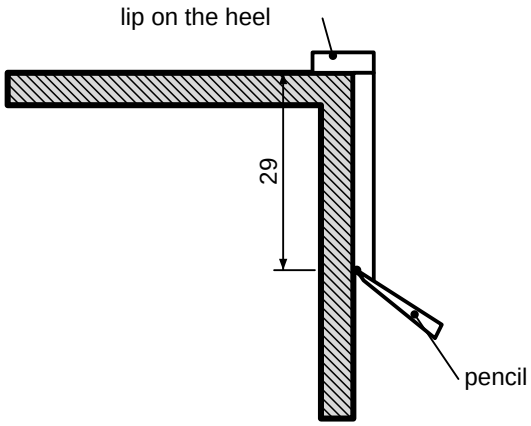
Fig. E-5 · One hook, every station. The station line only has to cross the leg; the jig finds the 19 mm.

3 **Pencil the two lines with the rail marking gauge, full length.** Screw line: lip over the flat leg's free edge, pencil in the **19** notch, slide. Bolt line: gauge on the vertical leg's outer face, lip over the heel, pencil in the **29** notch, slide. The bolt line is only a check: the fitting's holes must sit on it in step 6.

a SCREW LINE on the flat leg
lip over the free edge,
pencil in the 19 notch, slide



b BOLT LINE on the vertical leg
gauge on the outer face, lip over the heel,
pencil in the 29 notch, slide



Sections, marking gauge (P18) in use. Run each line the full length of both rails.

Fig. E-6 · The gauge's lip is the zero; the V notches put the pencil point on 19.0 and 29.0. SOURCE-DEFINED

4 **Pilot the screw holes through the drill jig.** Jig lip down over the flat leg's free edge, sight grooves on the station line, hold it down and drill 1/8 in through its bore: square, on the 19 mm line, no punch needed. All 19 stations (9 on rail A, 10 on rail B). Use a drill press if you have one.

5 **Open and deburr.** Jig off. Open every pilot to **⌀5.56 (7/32 in)** at low speed with a drop of oil on each; the bit follows the pilot. Deburr both faces with the file or a larger bit turned by hand.

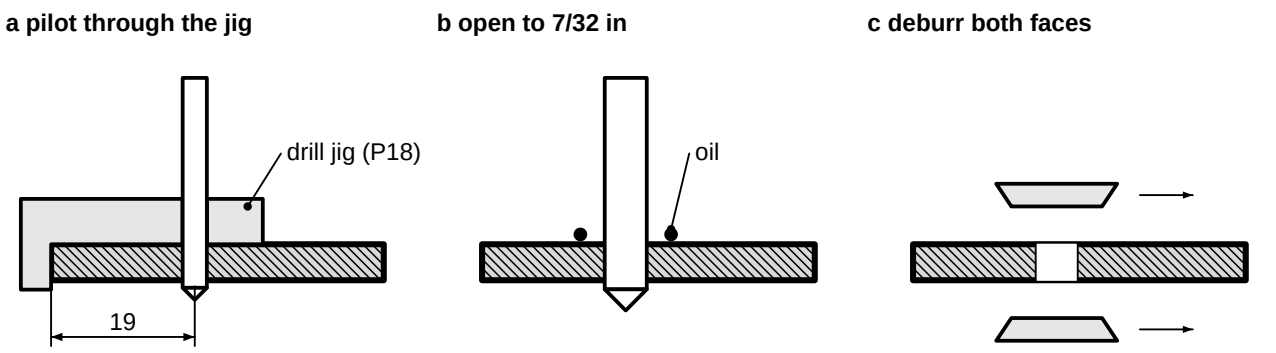


Fig. E-7 · Flat leg in section. The jig's bore is 1/8 in plus a running clearance; it keeps the bit square and on the 19 line.

THE FITTING IS THE TEMPLATE VERIFIED
The fittings were measured on 23 Oct 2026: legs 100.5 × 41, two 21 × 12 slots per leg with centres 29.15 and 78.45 mm from the heel. That puts the six bolt holes at the positions below. Even so, do not lay them out with the tape: clamp the fitting on its heel mark and drill through it with the printed **slot plug** (steps 6 and 7), so each hole lands on its slot's centre. Check first that a plug drops into each slot without rattle. The heel of a fitting is the **outside of its bend**.

Clamp the fitting	Heel on	Legs toward	Holes it makes (cm from the labelled end)
Rail A, outside fitting	the 52.0 cm mark	the far end	O1 54.9, O2 59.8
Rail A, inside fitting	the 51.5 cm mark	the wall end	I1 48.6, I2 43.7
Rail B, one fitting (both share the holes)	the corner end itself (0)	the far end	C1 2.9, C2 7.8

Heel stations SOURCE-DEFINED; hole positions from the measured slot centres VERIFIED. Both fittings are identical, so holes copied from one fit the other.

6 **Clamp the fitting and pilot through the plugs.** Rail with the vertical leg's outer face up. Clamp one fitting to the vertical leg with its heel exactly on the mark and its holes centred on the 29 mm line, legs the way Fig. E-8 shows. Drop a slot plug into each slot and drill the 1/8 in pilot through each plug's bore into the rail. Repeat for each of the three placements.

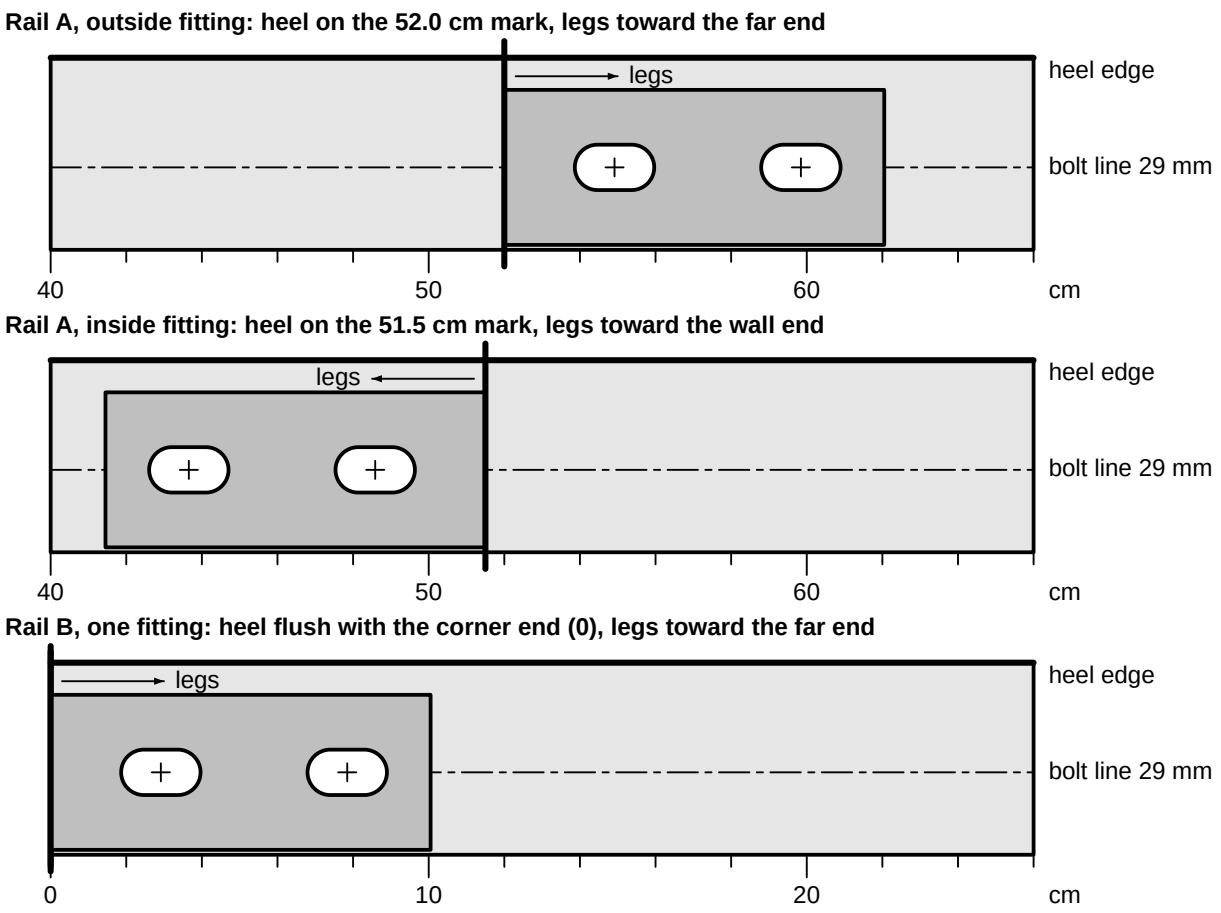


Fig. E-8 · The three clamp positions, vertical leg seen face on (1:2). Heavy line: the heel mark. Slots and their centres as measured. **VERIFIED**

7 **Fitting off, open to ⌀11.1 (7/16 in).** Cobalt bit, slow, oil on each. The bit follows the pilot, so the hole lands on the slot's centre, not anywhere along its 21 mm. Deburr. Six holes: four in rail A, two in rail B.

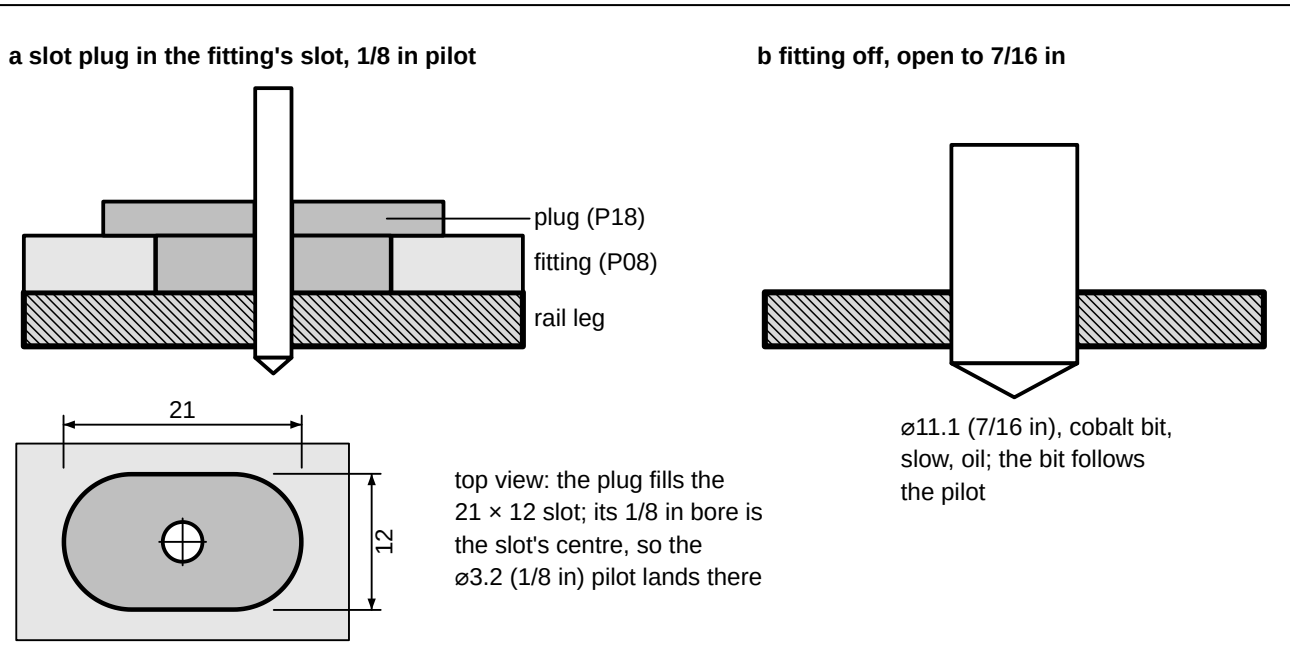
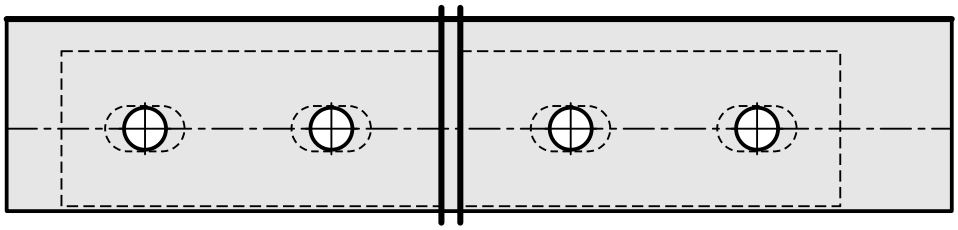


Fig. E-9 · The slot plug: a stadium that fills the 21 × 12 slot with a running clearance, 1/8 in bore at its centre. Without it a punch could land anywhere along the slot.

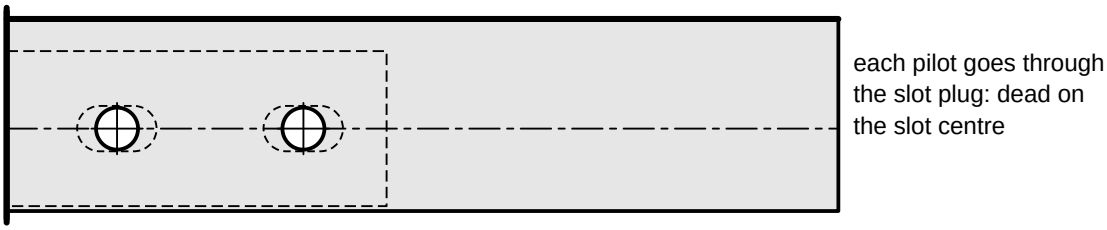
WHY THE HOLES ON BOTH RAILS MEET WITHOUT A JIG VERIFIED

1 · The same fitting is the template for all three placements: rail A heel on the 52.0 cm mark legs toward the far end, rail A heel on the 51.5 cm mark legs toward the wall end, rail B heel flush with the corner end.
2 · The slot plug puts every pilot on a slot centre, 29.15 and 78.45 mm from wherever the heel sat. So when a fitting is bolted on, its slots sit over the holes it made.
3 · Both fittings are identical, so the inside one fits the holes the outside one made.
4 · What makes the two rails meet at the corner is rail B's end against rail A's vertical leg with B's outer face on the 52.0 cm mark. The floor test (step 8) proves it before painting: every bolt must slide in by hand. If one won't, open that rail hole with a round file now, never under the desk.
The rail drill jig is only for the 19-line screw holes, on either rail; it plays no part in the bolt holes.

1 One fitting, three placements. Rail A, vertical leg:



Rail B, vertical leg:



2 Bolted: each slot sits over the hole it made (rail A from the knee space, rail B cut)

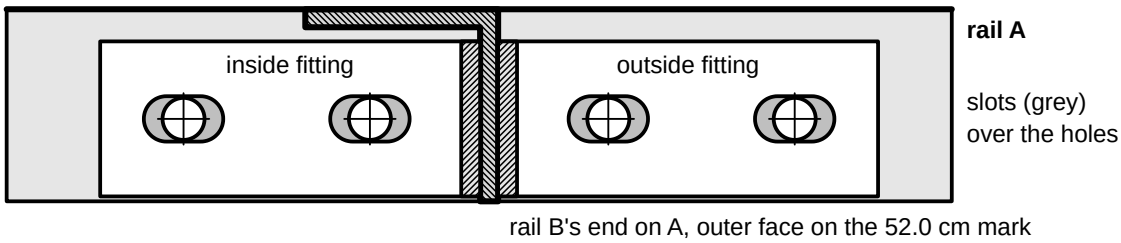
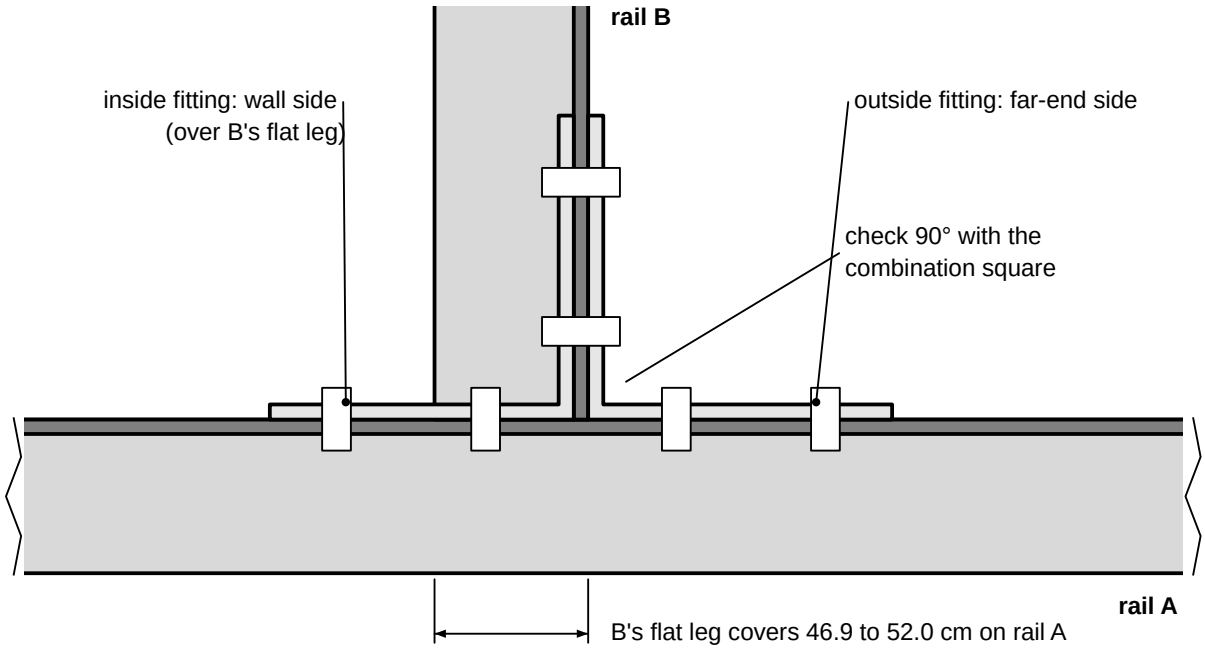


Fig. E-10 · Top: the one fitting in its three placements, heel marks heavy, slots dashed. Bottom: bolted at the corner, each slot over the hole it made. VERIFIED slot centres; SOURCE-DEFINED heel marks.

8 **Floor test.** Lay both rails upside down on the floor, flat legs down, vertical legs up. Butt rail B's corner end against the outer face of rail A's vertical leg so that rail B's outer face is on rail A's **52.0 cm** mark; rail B's flat leg then covers 46.9 to 52.0 cm on rail A's tape. Bolt both fittings on loosely: the outside one on the far-end side of rail B, the inside one on the wall side. Every bolt must slide in by hand. Check the corner with the combination square. If a bolt will not go in, open that hole with a round file now, not under the desk.

Plan from above. Both rails upside down on the floor: flat legs down, vertical legs up.
Bolts in loosely; every one must slide in by hand.



B's outer face on rail A's 52.0 cm mark; B's end touching A: no gap

Fig. E-11 · Floor test, plan from above. Bolts drawn as boxes. SOURCE-DEFINED

9 **Finish.** Take the fittings off. Degrease both rails, scuff with 120 grit, two coats of flat black enamel, 24 h to cure. The holes need no masking.

The rails are now ready for the tops: rail A goes on top A with its wall end at x = 1 cm (1 cm in from the taped end), rail B on top B with its corner end projecting 12 cm past the joint end. Chapter F bolts the corner once both tops stand.

F · Steel corner assembly

The two rails are joined at the inside corner by two gusseted strut fittings, one each side of rail B's end, bolted to the vertical legs. This bolted joint, not the joining plates, carries the inside corner of top B. Both tops are standing (build manual step 7); you work from below.

> **SEQUENCE: THE FITTINGS GO ON LAST** SOURCE-DEFINED
> Before the first fitting goes on, all of these must be done:
> 1 · The two surfaces are flush at the joint, leveled with the feet (never packed).
> 2 · Rail B's end touches the outer face of rail A's vertical leg with no gap, and rail B's outer face is on the 52.0 cm mark on rail A.
> 3 · Joining plates P1 and P2 are screwed across the joint. They align the tops; they do not carry the corner.
> 4 · Rail B's screws 1 and 2 are driven into top A. The inside fitting sits right under them and would block the driver.

	Part	Qty		Hardware	Qty
P06	Rail A, angle 213 cm, drilled (E)	1	**H08**	3/8-16 × 1½ in hex bolt, grade 5	6
P07	Rail B, angle 213 cm, drilled (E)	1	**H09**	3/8 flat washer, under each head and nut	12
P08	Gusseted 90° strut fitting, 5 mm, legs 100.5 × 41 mm, two 21 × 12 mm slots per leg VERIFIED	2	**H10**	3/8-16 nylon-insert lock nut	6

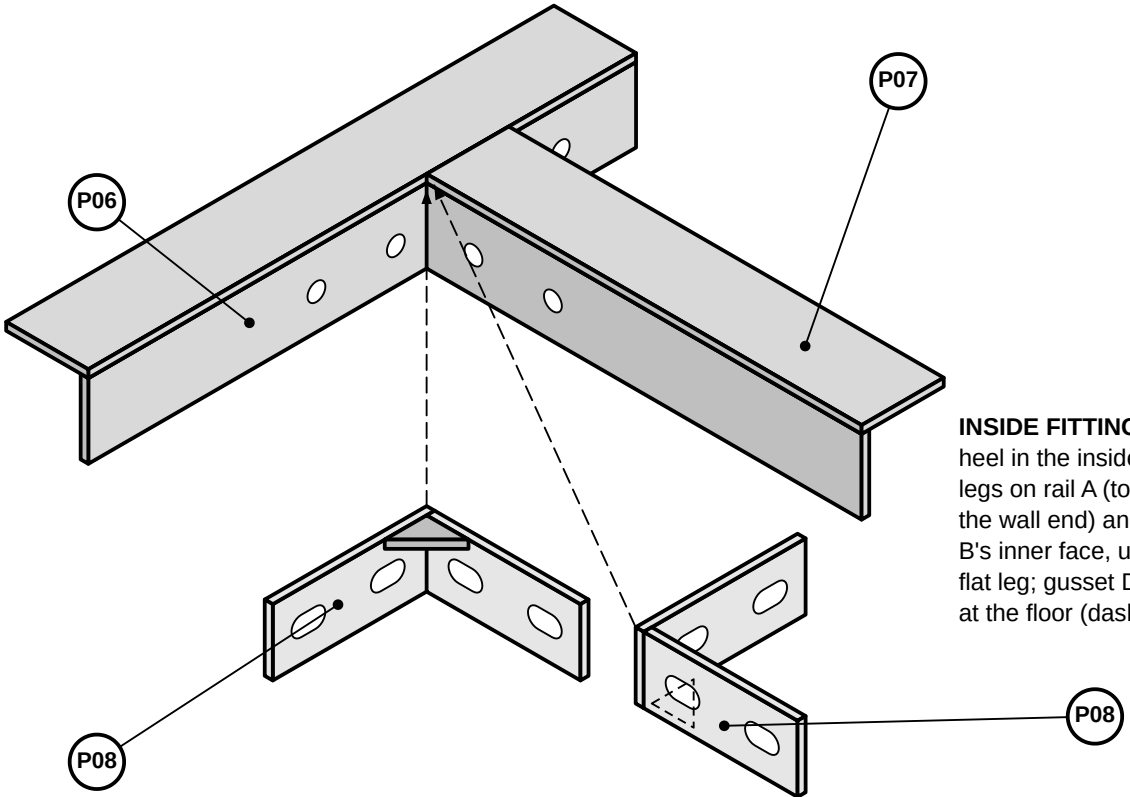


Fig. F-1 · The corner seen from the knee space with the tops lifted away. Both fittings are drawn pulled out of place; the dashed lines run from each heel to where it goes. The outside fitting fills the outside corner, the inside one the pocket under rail B's flat leg, gusset down. Neither is a mirror image: they are the same part, the inside one turned gusset-down. SOURCE-DEFINED placement.

Which bolt goes where

Bolts	Through	Head and washer	Washer and nut
O1, O2	outside fitting + rail A's vertical leg	fitting side, in the knee space	rail A's inner face, under its flat leg
I1, I2	inside fitting + rail A's vertical leg	fitting side, in the pocket under rail B	rail A's inner face, under its flat leg
C1, C2	outside fitting + rail B's vertical leg + inside fitting	outside fitting, knee side	inside fitting, in the pocket

Every bolt goes in **from the fitting side**: four through rail A's leg with one fitting each, two through rail B's leg with a fitting on each side.

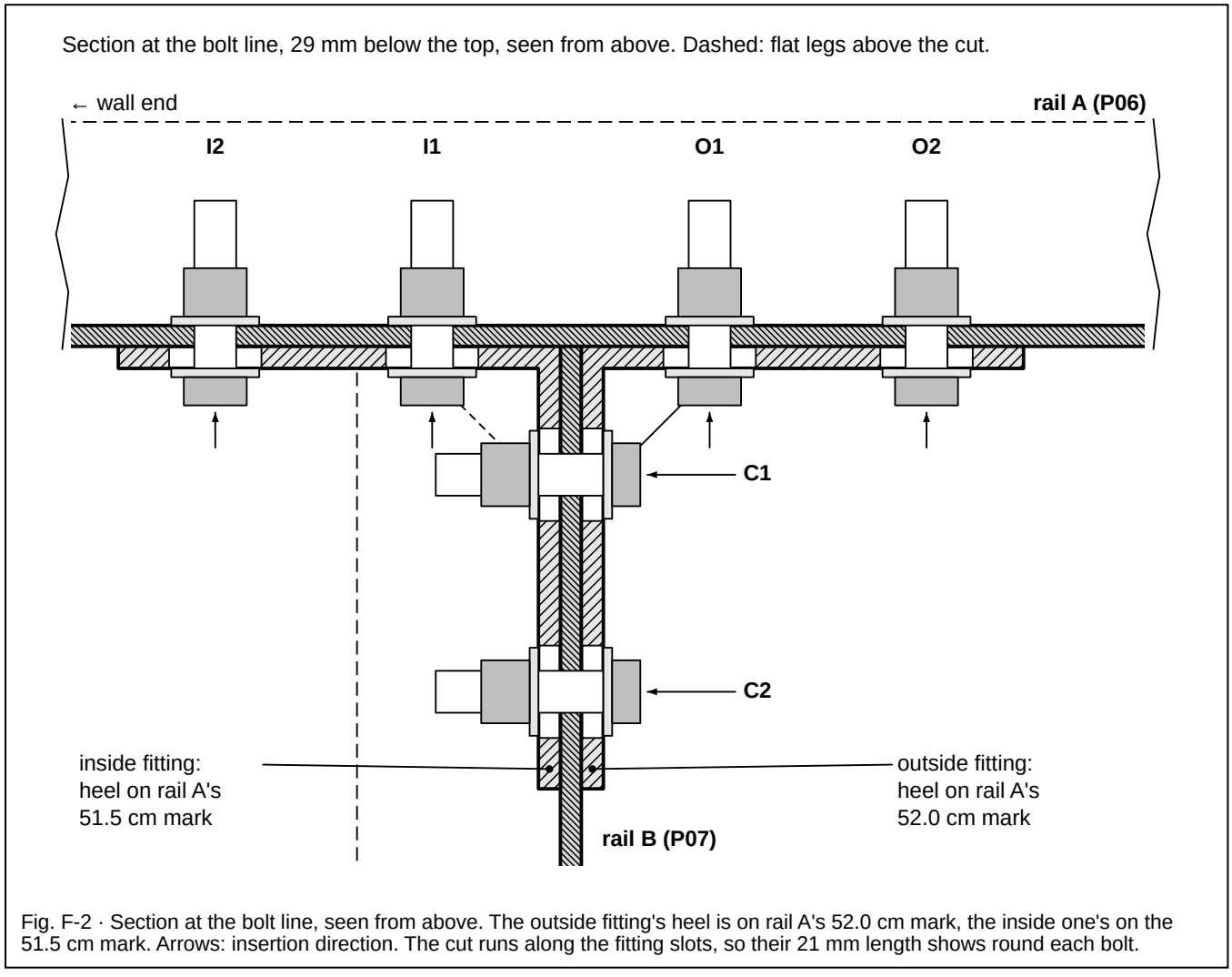


Fig. F-2 · Section at the bolt line, seen from above. The outside fitting's heel is on rail A's 52.0 cm mark, the inside one's on the 51.5 cm mark. Arrows: insertion direction. The cut runs along the fitting slots, so their 21 mm length shows round each bolt.

1

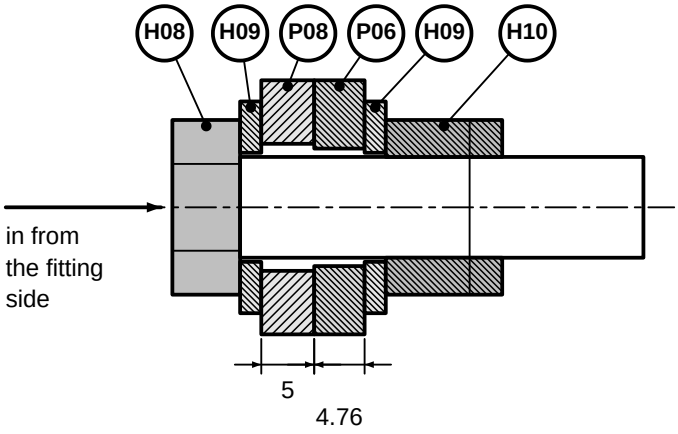
Outside fitting. Heel into the outside corner, one leg flat on rail A's vertical leg toward the far end, the other on rail B's outer face, gusset up, facing the knee space. Push O1 and O2 through from the fitting side with a washer under each head; washer and nut on rail A's inner face. Finger tight.

2

Inside fitting. Into the pocket: heel in the inside corner, one leg on rail A's vertical leg toward the wall end, the other on the inner face of rail B's vertical leg, **gusset pointing at the floor**. Gusset-up it would sit 1 mm under the head of rail B's screw 1. I1 and I2 from the fitting side, washers, nuts on rail A's inner face. Finger tight.

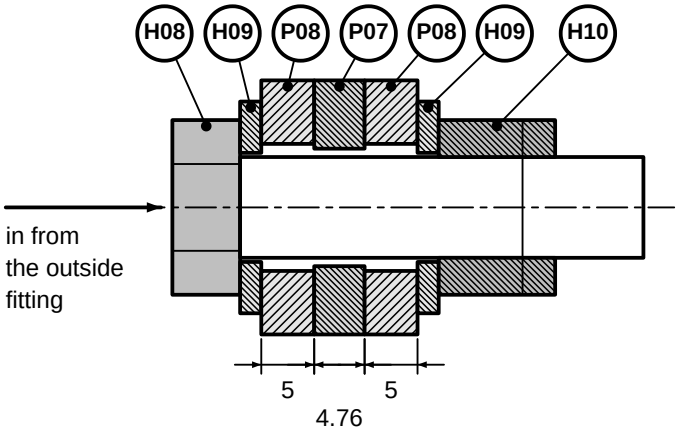
3 **Through bolts.** C1 and C2 from the outside fitting: through the outside fitting, rail B's vertical leg and the inside fitting, washer under the head, washer and nut in the pocket. Finger tight. All six bolts are now in; none is tight.

Rail A: 4 bolts, one fitting each (O1, O2, I1, I2)



fitting 21 × 12 slot (cut across): 2.5 mm play across, 11.5 mm along rail hole ⌀11.1 (7/16 in): 1.6 mm of play
so the joint holds by friction: tighten fully, 25 N·m

Rail B: 2 bolts through outside fitting, rail B and inside fitting (C1, C2)



head and washer on the outside fitting (knee side); washer and nut on the inside fitting, in the pocket under rail B's flat leg

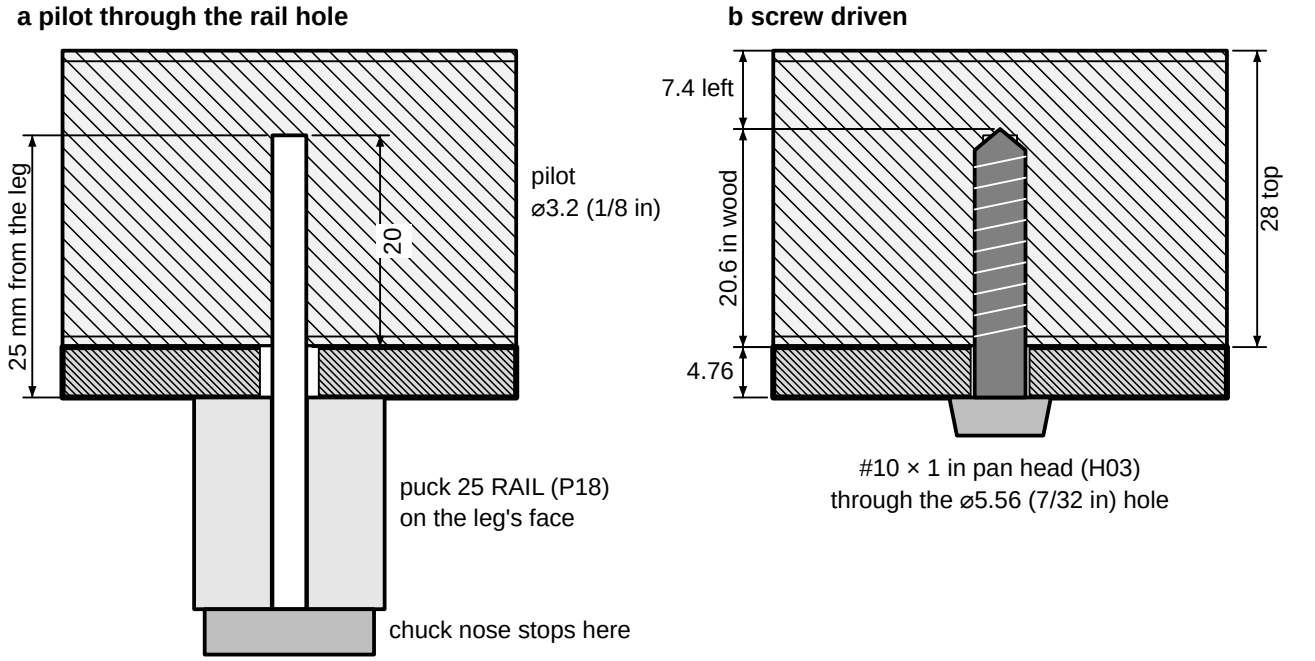
Fig. F-3 · Bolt stack-ups, cut along the bolt. All mm. Plates drawn cut off. The fitting's slots are 21 × 12; the cut shows their 12 width. VERIFIED fitting; SOURCE-DEFINED bolts.

4 **Check again, then tighten all six.** Fingertip across the joint: no step. Rail B's end still touching rail A. Then tighten every bolt fully:

! **TIGHTENING: 25 N·m (18 ft·lb)** SOURCE-DEFINED
With a torque wrench, **25 N·m (18 ft·lb)**, under the standard dry torque even for a grade 2 bolt. Without one, pull as hard as one hand can on a 15 cm wrench, then stop.
Why tight, not snug: this is a friction joint. The fittings' 21 × 12 slots leave 2.5 mm of play across and 11.5 mm along each bolt, the 7/16 in rail holes 1.6 mm, so only the clamping force keeps rail B's corner from settling.
When: only after the two surfaces are flush and rail B's end touches rail A.
Later: if the corner ever drops a millimetre, retighten these six bolts.

How the rails hold the tops

Each rail is screwed to its top through the flat leg: nine #10 × 1 in pan-head Robertson screws (H03) in rail A, ten in rail B, 19 in all. Rail B's holes 1 and 2 land in top A (its first 12 cm lies under top A) and are driven in build-manual step 7.4, before the fittings; the rest go in with the tops face down.



Never more than 21 mm into the 28 mm top. Drawn upright, as for rail B's holes 1 and 2.

Fig. F-4 · One rail screw in section. All mm. The pilot is 1/8 in, 25 mm from the flat leg's face (the face the puck stands on), which is 20 mm into the wood; the screw passes the 4.76 mm steel and goes 20.6 mm into the 28 mm top, leaving 7.4 mm under the laminate. SOURCE-DEFINED

Item	mm	Source
Top thickness	28	SOURCE-DEFINED MEASURE ON HARDWARE calipers (if it reads 27 mm, use screws 1 mm shorter everywhere in the top)
Flat leg	4.76	SOURCE-DEFINED
Screw into the top	20.6	SOURCE-DEFINED; limit 21 mm into any top
Pilot depth from the leg's face	25	printed drill guide, puck 25 RAIL SOURCE-DEFINED

Settling check. Lay a straightedge across the joint at the inside corner and write down any step. Check again after two weeks and after the first month; if the corner has settled, retighten the six fitting bolts.